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Array substrate and manufacturing method thereof

Abstract

An array substrate and a manufacturing method thereof are provided. The array substrate includes a substrate and a first recess. The first recess sequentially extends through a second inorganic layer, a third insulating layer, a first inorganic layer, a second insulating layer, a first insulating layer, a first semiconductor layer, and a portion of a barrier layer. A bottom surface of the first recess is formed inside the barrier layer.

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Background/Summary

FIELD OF DISCLOSURE
(1) The present disclosure relates to the field of display panels, in particular, relates to an array substrate and a manufacturing method thereof.
BACKGROUND
(2) Low temperature polycrystalline oxide (LTPO) is an organic light-emitting diode (OLED) display technology with low power consumption. LTPO thin film transistors (TFTs) have a lower driving power than low temperature poly-silicon (LTPS) TFTs. LTPS needs 60 Hz to display still images, but LTPO can be reduced to only 1 Hz, which greatly reduces a driving power. The LTPO converts part of transistors into oxides, so that there is less leakage current, and a capacitor voltage (charge) can be maintained for one second to drive 1 Hz. A leakage current of the LTPS is larger, even driving a stationary pixel also requires 60 Hz. Otherwise, a brightness will be greatly reduced, while the LTPO will not. Therefore, LTPO products with lower power consumption are increasingly sought after by people.
(3) In comparison with the LTPS, manufacturing the LTPO requires more layers. The LTPS transistor is manufactured, and then metal oxide (e.g., IGZO) is deposited on it by sputtering to form an oxide transistor. In order to prevent the LTPS transistor and the IGZO transistor from interfering with each other, there is a thicker inorganic film between the two as a barrier. Thus, there is a thicker inorganic film layer between a source/drain electrode (SD) and a semiconductor layer (poly). Due to a requirement of high pixels per inch (PPI), a diameter of the source/drain electrode connected to the semiconductor layer is usually small, generally between 2 μm to 4 μm, and a depth of a hole from the source/drain electrode to the semiconductor layer ranges from 1 μm to 1.5 μm. The narrow and deep hole is prone to following two defects during etching. The inorganic layer is easy to remain, and the remaining inorganic layer will increase a contact resistance when connecting the SD and the poly. In the deep and small hole, a wiring of the SD is easy to break, causing the SD and the poly to connect abnormally. The above problems can usually lead to a decrease in a yield of LTPO products.
SUMMARY OF DISCLOSURE
(4) The present disclosure provides an array substrate and a manufacturing method thereof to solve the technical problems that a contact hole of a source/drain electrode and an semiconductor layer in the array substrate is too small, which causes the source/drain electrode to break.
(5) The technical solutions to solve the above technical problems are as follows. The present disclosure provides an array substrate, including a substrate, a first recess, a third via hole, and a first source/drain electrode. A barrier layer, a first semiconductor layer, at least one insulating layer, at least one inorganic layer, and a second semiconductor layer are disposed on a surface of the substrate. The first recess extends through the at least semiconductor layer and a portion of the

barrier layer. A bottom surface of the first recess is formed inside the barrier layer. The third via hole is connected to a part of an upper surface of the second semiconductor layer. The first source/drain electrode covers an inner sidewall and the bottom surface of the first recess and an inner sidewall and the bottom surface of the third via hole. The first source/drain electrode is connected to the first semiconductor layer and the second semiconductor.

(6) Furthermore, the at least one insulating layer comprises a first insulating layer, a second insulating layer and a third insulating layer, and the at least one inorganic layer comprises a first inorganic layer and a second inorganic layer.

(7) Furthermore, the barrier layer, the first semiconductor layer, the first insulating layer, the second insulating layer, the first inorganic layer, the second semiconductor layer, the third insulating layer and the second inorganic layers are sequentially disposed.

(8) Furthermore, the first recess sequentially extends through the second semiconductor layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer and a portion of the barrier layer.

(9) Furthermore, the third via hole sequentially extends through the second inorganic layer and a portion of the third insulating layer.

(10) Furthermore, the array substrate further includes a first gate, a second gate, and a passivation layer. The first gate is disposed on a surface of the first insulating layer away from the barrier layer. The second insulating layer covers the first gate. The second gate is disposed on a surface of the second insulating layer away from the first insulating layer. The first inorganic layer covers the second gate.

(11) Furthermore, the passivation layer is disposed on the surface of the second inorganic layer away from the third insulating layer, covers the first source/drain electrode, and fills the first recess.

(12) Furthermore, the array substrate further includes a second recess and a first planarization layer. The second recess sequentially extends through the passivation layer, the second inorganic layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer, the barrier layer, and a portion of the substrate. The first planarization layer is disposed on a surface of the passivation layer away from the second inorganic layer and fills the second recess.

(13) Furthermore, the third via hole is located between the first recess and the second recess.

(14) Furthermore, the substrate includes a first flexible layer, a first buffer layer, a second flexible layer, and a second buffer layer. The first buffer layer is disposed on a surface of the first flexible layer. The second flexible layer is disposed on a surface of the first buffer layer away from the first flexible layer. The second buffer layer is disposed on a surface of the second flexible layer away from the first buffer layer.

(15) A bottom surface of the second recess coincides with a surface of the second flexible layer away from the first flexible layer.

(16) Furthermore, the first recess includes a first via hole and a first inverted trapezoid recess. The first via hole extends through the second inorganic layer and the third insulating layer. The first inverted trapezoid recess extends through the first inorganic layer, the second insulating layer, the first insulating layer, the first semiconductor layer, and a portion of the barrier layer. The first inverted trapezoid recess is disposed correspondingly to the first via hole, and a width of an opening of the first inverted trapezoid recess is less than a diameter of the first via hole.

(17) Furthermore, an included angle between the inner sidewall of the first via hole and a surface of the first inorganic layer ranges from 40 degrees to 80 degrees.

(18) The present disclosure also provides a manufacturing method of an array substrate, including following steps: providing a substrate; forming a barrier layer, a first semiconductor layer, at least one insulating layer, at least one inorganic layer, and a second semiconductor layer on the substrate; etching the at least one inorganic layer, the at least one insulating layer, the first semiconductor layer, and a portion of the barrier layer to form a first recess, where a bottom surface of the first recess is formed inside the barrier layer; etching the at least one inorganic and a portion of the

barrier layer to form a third via hole, the third via hole being connected to a part of an upper surface of the second semiconductor layer; forming a first source/drain electrode on the at least one inorganic layer at a top edge of the first recess, and covering an inner sidewall and the bottom surface of the first recess and an inner sidewall and the bottom surface of the third via hole with a middle section of the first source/drain electrode; and forming a passivation layer on the second inorganic layer, where the passivation layer covers the first source/drain electrode.
(19) Furthermore, the at least one insulating layer comprises a first insulating layer, a second insulating layer and a third insulating layer, and the at least one inorganic layer comprises a first inorganic layer and a second inorganic layer.
(20) Furthermore, the barrier layer, the first semiconductor layer, the first insulating layer, the second insulating layer, the first inorganic layer, the second semiconductor layer, the third insulating layer and the second inorganic layers are sequentially disposed.
(21) Furthermore, the first recess sequentially extends through the second semiconductor layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer and a portion of the barrier layer.
(22) Furthermore, the third via hole sequentially extends through the second inorganic layer and a portion of the third insulating layer.
(23) Furthermore, after forming the first insulating layer, the method further includes a step of forming a first gate on the first insulating layer. The first gate corresponds to the first semiconductor layer. After forming the second insulating layer, the method further includes a step of forming a second gate on the second insulating layer. The second gate corresponds to the first gate.
(24) Furthermore, the step of forming the first recess includes: etching the second inorganic layer and the third insulating layer to form a first via hole; and etching the first inorganic layer, the second insulating layer, the first insulating layer, the first semiconductor, and a portion of the barrier layer at a position corresponding to a second via hole to form a first inverted trapezoid recess. A stepped surface is formed between a sidewall of the first via hole and a sidewall of the first inverted trapezoid recess, and the stepped surface is formed on the first inorganic layer.
(25) Furthermore, when the first via hole is formed, the second inorganic layer and the third insulating layer are etched to form the second via hole stimulatingly. When the first inverted trapezoid recess is formed, the first inorganic layer, the second insulating layer, the first insulating layer, the barrier layer, and a portion of the substrate at a position corresponding to the second via hole are etched to form a second inverted trapezoid recess, and the second inorganic layer and a portion of the third insulating layer are etched to form the third via hole stimulatingly, and the second via hole and the second inverted trapezoid recess are connected to form a second recess. The first via hole, the second via hole and the third via hole are formed by a same mask, and the first inverted trapezoid recess and the second inverted trapezoid recess are formed by a same mask.
(26) Furthermore, the third via hole is located between the first recess and the second recess.
(27) Advantages of the present disclosure are as follow. In the array substrate and the manufacturing method thereof, the sidewall of the first recess extends through the first semiconductor and the source/drain electrode. By connecting the first recess with the first semiconductor, a contact area of the first semiconductor and the source/drain electrode is increased, and a contact resistance is reduced. The stepped surface is formed on the sidewall of the first recess, and the source/drain electrode covers the stepped surface, which has a buffering effect and prevents the source/drain electrode from breaking caused by an excessively long side wall of the first recess.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) With reference to the accompanying drawings, through a detailed description of the specific embodiments of the present disclosure, the technical solutions and other beneficial effects of the present disclosure will be apparent.

(2) FIG. **1** is a schematic diagram of an array substrate of an embodiment.

(3) FIG. **2** is a schematic diagram of a first source/drain electrode of the embodiment.

(4) FIG. **3** is a schematic diagram of a first via hole of the embodiment.

(5) FIG. **4** is a schematic diagram of a first recess of the embodiment.

(6) Reference numerals in the drawings are as follows. substrate **1**; barrier layer **2**; first semiconductor layer **31**; first insulating layer **4**; first gate **5**; second insulating layer **6**; second gate **7**; first inorganic layer **8**; second semiconductor layer **32**; third insulating layer **9**; second inorganic layer **10**; first source/drain electrode **11**; passivation layer **12**; first planarization layer **13**; second source/drain electrode **14**; second planarization layer **15**; anode layer **16**; pixel definition layer **17**; luminescent layer **18**; stepped surface **121**; first recess **100**; first via hole **110**; first inverted trapezoid recess **120**; second recess **200**; second via hole **210**; second inverted trapezoid recess **220**.

DETAILED DESCRIPTION

(7) The technical solutions in the embodiments of the present disclosure will be clearly and completely described below in conjunction with the drawings of the embodiments of the present disclosure. Apparently, the described embodiments are only a part of the embodiments of the present disclosure, rather than all the embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative efforts are within the protection scope of the present disclosure.

EMBODIMENTS

(8) As shown in FIG. **1**, in this embodiment, an array substrate of the present disclosure includes a substrate **1**, a barrier layer **2**, a first semiconductor layer **31**, a first insulating layer **4**, a first gate **5**, a second insulating layer **6**, a second gate **7**, a first inorganic layer **8**, a second semiconductor layer **32**; a third insulating layer **9**, a second inorganic layer **10**, a first source/drain electrode **11**, a passivation layer **12**, a first planarization layer **13**, a second source/drain electrode **14**, a second planarization layer **15**, an anode layer **16**, a pixel definition layer **17**, and a luminescent layer **18**.

(9) The substrate **1** is a flexible substrate, and includes a first flexible layer **101**, a first buffer layer **102**, a second flexible layer **103**, and a second buffer layer **104**. The first flexible layer **101** and the second flexible layer **103** are disposed layer-by-later and have good flexibility, which is convenient for bending and folding the array substrate. The first buffer layer **102** and the second buffer layer **104** arranged between the flexible layers can well protect the first flexible layer **101** and the second flexible layer **103** and avoid cracks due to excessive stress during a bending process.

(10) The barrier layer **2** is disposed on a surface of the substrate **1**. The barrier layer **2** is configured to prevent external moisture from entering the array substrate through the substrate **1**, so as to prevent the moisture from corroding a circuit structure inside the array substrate and increase lifespan of the array substrate.

(11) The first semiconductor layer **31** is disposed on an upper surface of the barrier layer **2**. The first semiconductor layer **31** is made of a semiconductor material. The semiconductor material includes indium gallium zinc oxide (IGZO), indium gallium titanium oxide (IZTO), and indium gallium zinc titanium oxide (IGZTO). A thickness of the first semiconductor layer **31** ranges from 100 angstroms to 1000 angstroms. The first semiconductor layer **31** provides circuit support for the display panel.

(12) The first insulating layer **4** is disposed on a surface of the barrier layer **2** away from the substrate **1**. The first insulating layer **4** is made of an inorganic material. The inorganic material includes silicon oxide, silicon nitride, or a multilayer film structure. A thickness of the first insulating layer **4** ranges from 1000 angstroms to 3000 angstroms. The first insulating layer **4**

corresponds to the first semiconductor layer **31** and covers the first semiconductor layer **31**. The first insulating layer **4** functions as an insulation and is configured to prevent short circuits between lines inside the array substrate.

(13) The first gate **5** is disposed on an upper surface of the first insulating layer **4**. The first gate **5** is made of a metal material. The metal material includes molybdenum (Mo), aluminum (Al), copper (Cu), titanium (Ti), etc., an alloy, or a multilayer film structure. A thickness of the first gate **5** ranges from 2000 angstroms to 8000 angstroms. The first gate **5** and the first insulating layer **4** are arranged correspondingly.

(14) The second insulating layer **6** is disposed on a surface of the first insulating layer **4** away from the buffer layer **2**. The second insulating layer **6** is made of an inorganic material. The inorganic material includes silicon oxide, silicon nitride, or a multilayer film structure. A thickness of the second insulating layer **6** ranges from 1000 angstroms to 3000 angstroms. The second insulating layer **6** corresponds to the first semiconductor layer **31** and covers the first semiconductor layer **31**. The second insulating layer **6** functions as an insulation and is configured to prevent short circuits between the lines inside the array substrate.

(15) The second gate **7** is disposed on a surface of the second insulating layer **6** away from the first insulating layer **4**. The second gate **7** is made of metal material. The metal material includes molybdenum (Mo), aluminum (Al), copper (Cu), titanium (Ti), etc., or an alloy, or a multilayer film structure. A thickness of the second gate **7** ranges from 2000 angstroms to 8000 angstroms. The second gate **7** corresponds to the second insulating layer **6**.

(16) The first inorganic layer **8** is disposed on upper surfaces of the second gate **7** and the second insulating layer **6**. The first inorganic layer **8** is an interlayer insulating layer. The first inorganic layer **8** is made of an inorganic material. The inorganic material includes silicon oxide, silicon nitride, or a multilayer film structure. The first inorganic layer **8** functions as an insulation and is configured to prevent short circuits. A thickness of the first dielectric layer **8** ranges from 2000 angstroms to 10000 angstroms.

(17) The third insulating layer **9** is disposed on a surface of the first inorganic layer **8** away from the second insulating layer **6**. The third insulating layer **9** is made of an inorganic material. The inorganic material includes silicon oxide, silicon nitride, or a multilayer film structure. A thickness of the first insulating layer **4** ranges from 1000 angstroms to 3000 angstroms. The third insulating layer **9** corresponds to the first semiconductor layer **31** and covers the first semiconductor layer **31**. The third insulating layer **9** functions as an insulation and is configured to prevent short circuits between the lines inside the array substrate.

(18) The second inorganic layer **10** is disposed on a surface of the third insulating layer **9** away from the first inorganic layer **8**. The second inorganic layer **10** is an interlayer insulating layer. The second inorganic layer **10** is made of an inorganic material. The inorganic material includes silicon oxide, silicon nitride, or a multilayer film structure. The second inorganic layer **10** functions as an insulation and is configured to prevent short circuits. A thickness of the second inorganic layer **10** ranges from 2000 angstroms to 10000 angstroms.

(19) In order to facilitate an electrical connection between the first source/drain electrode **11** and the first semiconductor layer **31**, it is generally necessary to form via holes on the second inorganic layer **10**, the third insulating layer **9**, the first inorganic layer **8**, the second insulating layer **6**, and the first insulating layer **4**. The first semiconductor layer **31** is partially exposed by the via holes, so that the first source/drain electrode **11** is electrically connected to the first semiconductor layer **31** through the via holes.

(20) In this embodiment, a top surface of the first recess **100** coincides with a surface of the second dielectric layer **10** away from the third insulating layer **9**, and its sidewall extends through the second inorganic layer **10**, the third insulating layer **9**, the first inorganic layer **8**, the second insulating layer **6**, the first insulating layer **4**, the first semiconductor layer **31**, and a portion of the barrier layer **2**. A bottom surface of the first recess **100** is formed inside of the barrier layer **2**. The

first recess **100** extends through the first semiconductor layer **31**, that is, the first semiconductor layer **31** is partially exposed on a sidewall of the first recess **100**.
(21) As shown in FIG. **2**, both ends of the first source/drain electrode **11** are formed on the second inorganic layer **10** at a top edge of the first recess. A middle section of the first source/drain electrode **11** cover the sidewall and the bottom surface of the first recess **100**.
(22) Since the first semiconductor layer **31** is partially exposed on the sidewall of the first recess **100**, the first source/drain electrode **11** covers the sidewall of the first recess **100**, and the first source/drain electrode **11** is electrically connected to the first semiconductor layer **31**. Also, the bottom surface of the first recess **100** is formed inside the barrier layer **2**, so the first source/drain electrode can be completely connected to a cross-sectional surface of the first semiconductor layer **31**, thereby improving transmission performance.
(23) As shown in FIG. **3** and FIG. **4**, in order to better protect the first source/drain electrode **11**, specifically, the first recess **100** includes a first via hole **110** and a first inverted trapezoid recess **120**. A top surface of the first via hole **110** is flush with the surface of the second inorganic layer **10** away from the third insulating layer **9**. The first via hole **110** extends through the second inorganic layer **10** and the third insulating layer **9**. A bottom surface of the first via hole **110** is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**.
(24) A top surface of the first inverted trapezoid recess **120** is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**. Its sidewall extends through the first inorganic layer **8**, the second insulating layer **6**, the first insulating layer **4**, the first semiconductor layer **31**, and a portion of the barrier layer **2**. A bottom surface of the first inverted trapezoid recess **120** is formed inside the barrier layer **2**.
(25) An area of the bottom surface of the first via hole **110** is greater than an area of the top surface of the first inverted trapezoid recess **120**. That is, a stepped surface **121** is formed between the bottom surface of the first via hole **110** and the top surface of the first inverted trapezoid recess **120**. The stepped surface **121** is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**.
(26) When the first source/drain electrode **11** covers the inner wall of the first recess **100**, the stepped surface **121** reduces a tilting force. The stepped surface **121** can act as a buffer to prevent the first source/drain electrode **11** from breaking due to excessively long side walls.
(27) In this embodiment, an inner included angle between the sidewall of the first via hole **110** and the bottom surface ranges from 40 degrees to 80 degrees. An internal tension of the first source/drain electrode **11** can be well reduced, and the problem of the first source/drain electrode **11** breaking inside the first recess **100** can be avoided.
(28) The passivation layer **12** is disposed on a surface of the second inorganic layer **10** away from the third insulating layer **9**. Material of the passivation layer **12** includes SiOx, SiNx, AlO, and a stack of SiO and SiN. A thickness of the passivation layer **12** ranges from 100 nm to 1000 nm. The passivation layer **12** has functions of insulating and isolating external moisture and oxygen, and can prevent hydrogen atoms or moisture from entering the first source/drain electrode **11**.
(29) In this embodiment, the array substrate includes a display area **21** and a non-display area **22**. The first semiconductor layer **31**, the first gate **5**, the second gate **7**, and the first source/drain electrode **11** correspond to the display area **21**. The substrate **1**, the barrier layer **2**, the first insulating layer **4**, the second insulating layer **6**, the first dielectric layer **8**, the third insulating layer **9**, the second dielectric layer **10**, and the passivation layer **12** extend from the display area **21** to the non-display area **22**.
(30) In the non-display area **22**, a top surface of the second recess **200** coincides with a surface of the passivation layer **12** away from the second inorganic layer **10**. Its sidewall extends through the passivation layer, the second inorganic layer **10**, the third insulating layer **9**, the first inorganic layer **8**, the second insulating layer **6**, the first insulating layer **4**, the barrier layer **2**, and the second buffer layer **104**. A bottom surface of the second recess **200** is flush with a surface of the second

flexible layer **103** away from the first buffer layer **102**.
(31) Specifically, the second recess **200** includes a second via hole **210** and a second inverted trapezoid recess **220**. A top surface of the second via hole **210** is flush with the surface of the passivation layer **12** away from the second inorganic layer **10**. The second via hole **110** extends through the passivation layer **12**, the second inorganic layer **10**, and the third insulating layer **9**. A bottom surface of the second via hole **210** is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**.
(32) A top surface of the second inverted trapezoid recess **120** is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**. Its sidewall extends through the first inorganic layer **8**, the second insulating layer **6**, the first insulating layer **4**, the barrier layer **2**, and the second buffer layer **104**. A bottom surface of the second via hole **210** is flush with the surface of the second flexible layer **103** away from the first buffer layer **102**.
(33) An area of the bottom surface of the second via hole **210** is greater than an area of a top surface of the second inverted trapezoid recess **220**. That is, a stepped surface is formed between the bottom surface of the first via hole **110** and the top surface of the first inverted trapezoid recess **120**. The stepped surface is flush with the surface of the first inorganic layer **8** away from the second insulating layer **6**. An inner included angle between the inner sidewall and the bottom surface of the second via hole **210** ranges from 20 degrees to 70 degrees.
(34) The first planarization layer **13** is disposed on an upper surface of the passivation layer **12**. The first planarization layer **13** extends from the display area **21** to the non-display area **22**, and fills the second recess **200**.
(35) The second source/drain electrode **14** is disposed on an upper surface of the first planarization layer **13**, extends through the first planarization layer **13**, and is electrically connected to the first source/drain electrode **11**.
(36) The second planarization layer **15** is disposed on the upper surface of the first planarization layer **13**, and the second planarization layer **15** covers the second source/drain electrode **14**.
(37) The anode layer **16** is disposed on the second planarization layer **15**, extends through the second planarization layer **15**, and is electrically connected to the second source/drain electrode **14**.
(38) The pixel definition layer **17** is disposed on an upper surface of the second planarization layer **15**, and the pixel definition layer **17** covers the anode layer **16**. Pixel openings are formed in a region of the pixel definition layer **17** corresponding to the anode layer **16**. The pixel openings are filled with the luminescent layer **18**. The luminescent layer **18** is connected to the anode layer **16**.
(39) In order to better explain the present disclosure, an embodiment is also provided a manufacturing method of an array substrate, and the specific steps include the following.
(40) A substrate formation step S**1**:
(41) A substrate **1** is provided. The substrate **1** includes a first flexible layer **101**, a first buffer layer **102**, a second flexible layer **103**, and a second buffer layer **104** which are sequentially formed, and the substrate **1** has good flexibility.
(42) A thin film transistor formation step S**2**:
(43) A barrier layer **2** is formed on the substrate **1**.
(44) A semiconductor layer is formed on the barrier layer **2**. The semiconductor layer is active to form a semiconductor layer.
(45) A first insulating layer **4** is formed on the barrier layer **2**, and the first insulating layer **4** covers the first semiconductor layer **31**.
(46) A first gate **5** is formed on the first insulating layer **4**.
(47) A second insulating layer **6** is formed on the first insulating layer **4**, and the second insulating layer **6** covers the first gate.
(48) A second gate **7** is formed on the second insulating layer **6**.
(49) A first dielectric inorganic layer **8** is formed on the second insulating layer **6**, and the first inorganic layer **8** covers the second gate **7**.

(50) A third insulating layer **9** is formed on the first inorganic layer **8**.
(51) A second inorganic layer **10** is formed on the third insulating layer **2**.
(52) A first recess formation step and a second recess formation step S**3**:
(53) The second inorganic layer **10** and the third insulating layer **2** are etched by a first mask.
(54) A first via hole **110** is formed correspondingly to a display area, and a second via hole **210** is formed correspondingly to a non-display area, further, the second inorganic layer **10** and a portion of the third insulating layer **9** are etched by the first mask to form the third via hole **310**. The first via hole is a cylindrical via hole, and the second via hole **210** is a long rectangular via hole.
(55) The first inorganic layer **8**, the second insulating layer **6**, the first insulating layer **4**, the first semiconductor layer **31**, and a portion of the barrier layer **2** are etched by a second mask. A first inverted trapezoid recess is formed correspondingly to the first via hole **110**. The first via hole **110** is connected to the first inverted trapezoid recess to form a first recess **100**. A second inverted trapezoid recess is formed correspondingly to the second via hole **210**. The second via hole **210** is connected to the second inverted trapezoid recess to form a second recess **200**.
(56) In this embodiment, the first via hole **110** and the second via hole **210** are formed by using the first mask, and the first inverted trapezoid recess and the second inverted trapezoid recess are formed by using the second mask. Thus, a number of masks is saved, process steps are also reduced, a manufacturing cost is reduced, and a product yield is improved.
(57) A stepped surface **121** is formed between a sidewall of the first via hole **110** and a sidewall of the first inverted trapezoid recess, and the stepped surface **121** is formed on the first dielectric inorganic layer **8**.
(58) A first source/drain electrode formation step S**4**:
(59) A first source/drain electrode is formed on an inorganic layer at a top edge of the first recess **100**. A middle section of the first source/drain electrode covers an inner sidewall and the bottom surface of the first recess **100**. The first source/drain electrode covers the stepped surface **121**, which is beneficial to reduce an internal tension of the first source/drain electrode and prevent the first electrode from breaking.
(60) A passivation layer formation step S**5**:
(61) A passivation layer **12** is formed on the second inorganic layer **10**. The passivation layer **12** covers the first source/drain electrode and fills the first recess **100** and the second recess **200**.
(62) A passivation layer etching step S**6**: the passivation layer **12** inside the second recess **200** is etched by a third mask.
(63) A planarization layer formation step S**7**:
(64) A planarization layer is formed on the passivation layer **12**. The planarization layer extends from the display area to a bonding area. The planarization layer fills the second recess **200**. In subsequent bending and bonding processes, the second recess **200** can reduce adverse effects of bending stress.
(65) Advantages of the present disclosure are as follow. In the array substrate and the manufacturing method thereof of the embodiments, the sidewall of the first recess extends through the second inorganic layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer, the first semiconductor layer, and a portion of the barrier layer. The first recess has a relatively large diameter, which can effectively prevent inorganic materials from remaining in the first recess and reduce the contact resistance of the first semiconductor layer and the source/drain electrode. The first recess makes the first semiconductor layer form a cross-sectional surface on the sidewall of the first recess, and the source/drain electrode is arranged along the sidewall of the first recess. By connecting the sidewall of the first recess with the first semiconductor layer, a contact area of the first semiconductor layer and the source/drain electrode is increased. Moreover, the stepped surface is formed on the inner sidewall of the first recess, which has a buffering effect and prevents the source/drain electrode from breaking caused by an excessively long side wall of the first recess.

(66) The description of the above embodiments is only used to help understand the technical solutions of the present disclosure and its core idea. A person of ordinary skill in the art should understand that the technical solutions described in the foregoing embodiments may be modified, or some of the technical features may be equivalently replaced. These modifications or replacements will not cause the essence of the corresponding technical solutions to deviate from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. An array substrate, comprising: a substrate, wherein a barrier layer, a first semiconductor layer, at least one insulating layer, at least one inorganic layer, and a second semiconductor layer are disposed on a surface of the substrate, and the first semiconductor layer and the second semiconductor layer are disposed in different layers; a first recess extending through at least the first semiconductor layer and a portion of the barrier layer, wherein a bottom surface of the first recess is formed inside the barrier layer; a third via hole connected to a part of an upper surface of the second semiconductor layer; and a first source/drain electrode covering an inner sidewall and the bottom surface of the first recess and an inner sidewall and the bottom surface of the third via hole, wherein the first source/drain electrode is connected to the first semiconductor layer and the second semiconductor.
2. The array substrate according to claim 1, wherein the at least one insulating layer comprises a first insulating layer, a second insulating layer and a third insulating layer, and the at least one inorganic layer comprises a first inorganic layer and a second inorganic layer.
3. The array substrate according to claim 2, wherein the barrier layer, the first semiconductor layer, the first insulating layer, the second insulating layer, the first inorganic layer, the second semiconductor layer, the third insulating layer and the second inorganic layers are sequentially disposed.
4. The array substrate according to claim 2, wherein the first recess sequentially extends through the second semiconductor layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer and a portion of the barrier layer.
5. The array substrate according to claim 2, wherein the third via hole sequentially extends through the second inorganic layer and a portion of the third insulating layer.
6. The array substrate according to claim 2, further comprising: a first gate disposed on a surface of the first insulating layer away from the barrier layer, wherein the second insulating layer covers the first gate; a second gate disposed on a surface of the second insulating layer away from the first insulating layer, wherein the first inorganic layer covers the second gate; and a passivation layer disposed on the surface of the second inorganic layer away from the third insulating layer, covering the first source/drain electrode, and filling the first recess.
7. The array substrate according to claim 6, further comprising: a second recess sequentially extending through the passivation layer, the second inorganic layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer, the barrier layer, and a portion of the substrate; and a first planarization layer disposed on a surface of the passivation layer away from the second inorganic layer and filling the second recess.
8. The array substrate according to claim 7, wherein the third via hole is located between the first recess and the second recess.
9. The array substrate according to claim 7, wherein the substrate comprises: a first flexible layer, a first buffer layer disposed on a surface of the first flexible layer; a second flexible layer disposed on a surface of the first buffer layer away from the first flexible layer; and a second buffer layer disposed on a surface of the second flexible layer away from the first buffer layer; and wherein a bottom surface of the second recess coincides with a surface of the second flexible layer away from the first flexible layer.

10. The array substrate according to claim 2, wherein the first recess comprises: a first via hole extending through the second inorganic layer and the third insulating layer; and a first inverted trapezoid recess extending through the first inorganic layer, the second insulating layer, the first insulating layer, the first semiconductor layer, and a portion of the barrier layer; and wherein the first inverted trapezoid recess is disposed correspondingly to the first via hole, and a width of an opening of the first inverted trapezoid recess is less than a diameter of the first via hole.
11. The array substrate according to claim 10, wherein an included angle between the inner sidewall of the first via hole and a surface of the first inorganic layer ranges from 40 degrees to 80 degrees.
12. A manufacturing method of an array substrate, comprising following steps: providing a substrate; forming a barrier layer, a first semiconductor layer, at least one insulating layer, at least one inorganic layer, and a second semiconductor layer on the substrate, and the first semiconductor layer and the second semiconductor layer being disposed in different layers; etching the at least one inorganic layer, the at least one insulating layer, the first semiconductor layer, and a portion of the barrier layer to form a first recess, wherein a bottom surface of the first recess is formed inside the barrier layer; etching the at least one inorganic and a portion of the barrier layer to form a third via hole, the third via hole being connected to a part of an upper surface of the second semiconductor layer; forming a first source/drain electrode on the at least one inorganic layer at a top edge of the first recess, and covering an inner sidewall and the bottom surface of the first recess and an inner sidewall and the bottom surface of the third via hole with a middle section of the first source/drain electrode; and forming a passivation layer, wherein the passivation layer covers the first source/drain electrode.
13. The manufacturing method of the array substrate according to claim 12, wherein the at least one insulating layer comprises a first insulating layer, a second insulating layer and a third insulating layer, and the at least one inorganic layer comprises a first inorganic layer and a second inorganic layer.
14. The manufacturing method of the array substrate according to claim 13, wherein the barrier layer, the first semiconductor layer, the first insulating layer, the second insulating layer, the first inorganic layer, the second semiconductor layer, the third insulating layer and the second inorganic layers are sequentially disposed.
15. The manufacturing method of the array substrate according to claim 13, wherein the first recess sequentially extends through the second semiconductor layer, the third insulating layer, the first inorganic layer, the second insulating layer, the first insulating layer and a portion of the barrier layer.
16. The manufacturing method of the array substrate according to claim 13, wherein the third via hole sequentially extends through the second inorganic layer and a portion of the third insulating layer.
17. The manufacturing method of the array substrate according to claim 13, wherein after forming the first insulating layer, the method further comprises following step: forming a first gate on the first insulating layer, wherein the first gate corresponds to the first semiconductor layer; and wherein after forming the second insulating layer, the method further comprises following step: forming a second gate on the second insulating layer, wherein the second gate corresponds to the first gate.
18. The manufacturing method of the array substrate according to claim 13, wherein the step of forming the first recess comprises: etching the second inorganic layer and the third insulating layer to form a first via hole; and etching the first inorganic layer, the second insulating layer, the first insulating layer, the first semiconductor layer, and a portion of the barrier layer at a position corresponding to a second via hole to form a first inverted trapezoid recess; and wherein a stepped surface is formed between a sidewall of the first via hole and a sidewall of the first inverted trapezoid recess, and the stepped surface is formed on the first inorganic layer.

19. The manufacturing method of the array substrate according to claim 18, wherein when the first via hole is formed, the second inorganic layer and the third insulating layer are etched to form the second via hole stimulatingly; when the first inverted trapezoid recess is formed, the first inorganic layer, the second insulating layer, the first insulating layer, the barrier layer, and a portion of the substrate at a position corresponding to the second via hole are etched to form a second inverted trapezoid recess, and the second inorganic layer and a portion of the third insulating layer are etched to form the third via hole stimulatingly, and the second via hole and the second inverted trapezoid recess are connected to form a second recess; and the first via hole, the second via hole and the third via hole are formed by a same mask, and the first inverted trapezoid recess and the second inverted trapezoid recess are formed by a same mask.

20. The manufacturing method of the array substrate according to claim 19, wherein the third via hole is located between the first recess and the second recess.